Vo Thanh Nguyen

AI Engineer

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Portfolio: vtnguyen04.github.io | GitHub: vtnguyen04 | LinkedIn

Objective

AI Engineer with hands-on experience in Machine Learning, Computer Vision, and AI pipeline automation. Proven track record in developing, optimizing, and deploying deep learning models for real-time edge environments and data workflows. Eager to engineer scalable, high-impact AI solutions for complex industrial and enterprise systems.

Technical Skills

Programming Languages: Python, C++, SQL, Bash

Machine Learning & Computer Vision: PyTorch, TensorFlow, CUDA, TensorRT (FP16/INT8), OpenCV, Hugging Face

Data Engineering & Automation: Pandas, NumPy, Scikit-learn, n8n, RESTful APIs, Webhooks

Deployment, Systems & Edge: Docker, Git, Linux SharedMemory, ZeroMQ, NVIDIA Jetson, Hailo NPU

Professional Experience

- **AI Researcher / Engineer** Dec 2024 – Present

  HCM-UTE Research Lab – *Advisor: Dr. Ha Duc Bui*

  - **Sim-to-Real Traffic Scene Understanding** (`Traffic-JEPA`) – *Co-First Author*:
    - Decoupled visual semantic understanding from caption generation under synthetic-to-real domain shift, resolving traffic queries into structured facts before generating descriptions.
    - Designed a VQA predictor on frozen V-JEPA 2.1 features with a training-free refinement module enforcing inter-question relationships and temporal event consistency.
    - Conditioned LoRA fine-tuned Qwen3-VL-8B on refined semantic facts, achieving 87.09% VQA accuracy on the official WTS benchmark ($S_2 = 60.0853$, 1st Worldwide, ECCV 2026 Workshop Oral).
  - **Generative Traffic Forecasting** (`JEPA-Guided-Diffusion`) – *Co-First Author*:
    - Engineered a decoupled traffic video forecasting framework bridging frozen V-JEPA 2.1 latent dynamics into a frozen Cosmos-Predict2.5 DiT video generator.
    - Trained a lightweight adapter via Hungarian set-matching loss to align predictive visual tokens with Cosmos cross-attention conditioning.
    - Developed a 5-level Laplacian pyramid filtering and WCT color transfer pipeline to eliminate video artifacts and trajectory drift (Score: 75.13, Rank 3, ECCV 2026 Workshop Poster).
  - **Bosch Future Mobility Challenge - Romania** – *Software Lead*:
    - Architected the full-stack autonomous driving software architecture for a 1:10 physical vehicle: real-time sensor fusion, lane tracking, and ZeroMQ IPC messaging.
    - Designed a dual-branch neural network resolving multi-task gradient conflict between 2D detection and trajectory planning with custom Frenet loss.
    - Quantized and deployed the model on NVIDIA Jetson Orin Nano via TensorRT INT8, benchmarking >100 FPS throughput at <25ms latency.

  Tech Stack: Python, C++, PyTorch, CUDA, TensorRT (FP16/INT8), V-JEPA, Cosmos-Predict2.5 DiT, Qwen3-VL, Jetson Orin Nano, ZeroMQ, Linux SharedMemory, Docker.

- **AI Automation Engineer** Aug 2025 – Nov 2025

  Learning Chain

  - Built n8n pipelines automating multi-source data crawling (Web, Tavily, Social) with deduplication into Google Sheets.
  - Automated end-to-end content creation and SEO publishing across Web, Facebook, and Telegram using OpenAI/Gemini APIs.
  - Implemented Pinecone vector indexing for document RAG and automated scheduled n8n workflow backups to GitHub.

  Tech Stack: n8n, OpenAI & Gemini APIs, Pinecone, Webhooks, Facebook Graph API, Telegram Bot, Git.

- **AI Engineer** Mar 2024 – Feb 2025

  AIoT & Semiconductor (`aiotsemi.vn`)

  - Deployed an industrial vision pipeline for PCB defect detection (Client: BACKER) with 98.2% accuracy.
  - Built a real-time driver monitoring system fusing 478-point face mesh, gaze vectors, and head pose.
  - Architected zero-copy IPC via Linux SharedMemory across TensorRT and ONNX backends (3.2× throughput).

  Tech Stack: Python, OpenCV, TensorRT, OpenVINO, ONNX Runtime, MediaPipe, Hailo NPU, SharedMemory, Docker.

Publications

• **Sim-to-Real Traffic Scene Understanding by Decoupling Semantics from Caption Generation with V-JEPA**
Nguyen Hoai Thuong Bui*, **Thanh Nguyen Vo***, Trinh Tra Giang Nguyen*, Ha Duc Bui†
ECCV 2026 Workshop (Oral Presentation) May 2026

• **JEPA Guided Diffusion: Predictive Vision-Language Conditioning for Generative Traffic Forecasting**
Trinh Tra Giang Nguyen*, **Thanh Nguyen Vo***, Nguyen Hoai Thuong Bui*, Ha Duc Bui†
ECCV 2026 Workshop (Poster Presentation) May 2026

• **An Integrated Perception-to-Planning Framework for Autonomous Driving using YOLOv11 and Frenet**
Thanh Nguyen Vo, Trinh Tra Giang Nguyen, Lu Anh Khoi Le, Ha Duc Bui†
IEEE GTSD 2026 Proceedings IEEE Xplore June 2026

* Equal contribution / Co-first authors † Corresponding author / Lab Advisor

Education

Ho Chi Minh City University of Technology and Engineering (HCMUTE) 2022 – 2026
Bachelor of Engineering in Robotics and Artificial Intelligence GPA: 3.28 / 4.0 | TOEIC: 735 / 990
Relevant Coursework: Artificial Neural Networks: 9.6/10, Computer Vision: 9.5/10, Artificial Intelligence: 9.1/10.
- **Graduation Thesis (Grade: 9.2/10):** *End-to-End Control System for Autonomous Vehicles*.

Honors & Awards

AI, Autonomous Systems & Data Competitions:
- **Champion (1st Worldwide, Track 2)** – NVIDIA AI City Challenge 2026
- **Top 3 Worldwide (Track 5)** – NVIDIA AI City Challenge 2026
- **Top 9 Finalist Worldwide (Best Team from Asia)** – Bosch Future Mobility Challenge, Romania 2025
- **150M VND Bosch Sponsorship** (Team "Drive Beyond") – Bosch Future Mobility Challenge 2026
- **Top 5 Finalist** – Datathon 2026: The Gridbreakers (VinUniversity) 2026

Competitive Programming & Algorithms:
- **Third Prize** – ICPC Vietnam National Programming Contest 2024
- **Bronze Medal** – Da Nang Code League 2024
- **Third Prize** – Mastering IT HCMUTE 2023
- **Second Prize (Specialized Division)** – Vietnamese Student Informatics Olympiad (OLP) 2023
- **Second Prize (Non-Specialized Division)** – Vietnamese Student Informatics Olympiad (OLP) 2022

Academic Honors:
- **Talent Scholarship** – HCMUTE Academic Merit Award 2022 – 2023

References

Dr. Ha Duc Bui – Head of Research Lab, HCMUTE buihaduc@hcmute.edu.vn